Cover Letter — Neurocomputing

14 July 2026

Dear Editors of *Neurocomputing*,

On behalf of my co-authors, I am pleased to submit our original research article, **“Memory-Bounded Online Learning under Structured Non-IID Streams: A Proxy-Based Sensitivity Study”**, for consideration in *Neurocomputing*.

What the paper does. The paper studies memory-bounded online learning under the structured non-IID conditions that real streams exhibit — Markov switching, seasonal drift, burst repetition, and long-range dependence — with an IID control. Three classical memory-bounded streaming learners (online SGD with feature hashing, Polyak–Ruppert averaged SGD, and a Count-Min-style hash classifier) are evaluated across all regimes with 10-seed means, 95% confidence intervals, and paired Wilcoxon tests, alongside a theory-inspired quantum performance proxy that serves as a forward-looking reference point. Two operational correlation parameters (refreshing time τ and repetition number r) organise the results into a sensitivity landscape; a two-dataset real-stream case study (NYC Taxi and Machine Temperature, Numenta Anomaly Benchmark) applies the full pipeline to real telemetry with imbalance-aware metrics (balanced accuracy, F1).

Fit to Neurocomputing. The paper engages the journal’s online-learning-under-drift line — including the drift-monitoring perspective of Hinder et al. (2024, *Neurocomputing*) and the broader online-learning survey of Hoi et al. (2021, *Neurocomputing*) — and contributes a regime-structured sensitivity methodology for assessing memory-bounded learners under correlation, together with fully reproducible baselines.

Declarations. The manuscript is original, unpublished, and not under consideration elsewhere. All authors approved the submission. No competing interests; no external funding. A generative-AI-assistance declaration is included per Elsevier policy. All code and data are open (Zenodo DOI: 10.5281/zenodo.19831893, MIT licence); every figure and table regenerates from seeded scripts.

Suggested reviewers: [to be entered by the authors in Editorial Manager].

Thank you for considering our work.

Sincerely, on behalf of all authors,

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